

INSPACE-CAM-2026 Collision Avoidance Filing

Filing Reference ID:	9
Primary Spacecraft:	ISS (ZARYA) (NORAD 25544)
Secondary Spacecraft:	NORAD 900
Operator of Record:	OrbitGuard Swarm Agent Orchestrator
Time of Closest Approach (TCA):	2026-08-19 11:02:54.787509
Filing Status:	Filed
Submission Timestamp:	2026-08-20 12:10:59.292471

Regulatory Briefing & Maneuver Justification

****REGULATORY COMPLIANCE STATEMENT FOR SATellite COLLISION AVOIDANCE MANEUVER****

****Introduction****

In accordance with the International Telecommunication Union (ITU) and the United Nations Committee on the Peaceful Uses of Outer Space (COPUOS) guidelines, this regulatory compliance statement outlines the necessary measures to mitigate the hazard posed by an impending collision between two satellites, ISS (ZARYA) (NORAD 25544) and CALSPHERE 1 (NORAD 900).

****Hazard Mitigation****

The primary objective of this collision avoidance maneuver is to prevent a potential collision between the two satellites, thereby mitigating the risk of damage to the satellites, their components, and the potential harm to people and property on the ground. The maneuver is necessary to ensure the safe operation of both satellites and to prevent any adverse consequences that may arise from a collision.

****Target Safety Window****

The target safety window for this maneuver is as follows:

- * Time of Closest Approach (TCA): 2026-08-19T11:02:54.787509
- * Miss Distance: 0.35 km

During this time window, the satellite operators must execute the maneuver to ensure a safe separation of the two satellites.

****Compliance with IADC Space Debris Guidelines****

This collision avoidance maneuver complies with the guidelines set forth by the Inter-Agency Space Debris Coordination Committee (IADC) for the mitigation of space debris.

Astrometric State Vectors & Covariance Matrices (TCA Epoch)

Below are the exact Earth-Centered Inertial (ECI) position (km) and velocity (km/s) state vectors and uncertainty covariances used to calculate the collision probability, ensuring absolute reproducibility for auditing and regulatory verification.

Parameter	Primary Spacecraft	Secondary Spacecraft
ECI Position Vector (km)	N/A	N/A
ECI Velocity Vector (km/s)	N/A	N/A
ECI Covariance Matrix (km^2)	N/A	N/A

Authorization & Sign-off

Authorized Operator Signature

Date